

ELECTRICITY BILLING SYSTEM

DOCUMENT TYPE

Project Statement

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1. Problem Statement

Core Challenge: Traditional manual electricity billing systems are highly prone to human error during data entry, validation anomalies (such as accepting current readings lower than prior entries), opaque tariff calculations, and delayed bill dispatch.

Utility providers require an automated, transparent, and modular computational system that validates meter inputs, seamlessly calculates progressive slab tariffs, accounts for mandatory fixed infrastructure charges and government taxes, and produces clean, standardized billing statements for end consumers.

2. Scope of the Project

The **Smart Electricity Billing System** is scoped as a modular Python-based computational framework designed for core utility calculation workflows. The boundaries of the current project release include:

In-Scope Functional Area

Meter data validation logic, progressive tier slab charge evaluation, mandatory fixed charge addition, dynamic 5% tax computing, and formatted terminal/print bill rendering.

Modular Design Scope

Separation of concerns into specialized modules (`calculator.py` for math logic, `bill.py` for visual display, and `final.py` for orchestration).

Out of Scope for Initial Release: Direct hardware meter reading collection via IoT sensors, payment gateway integration, and multi-tenant database persistence.

3. Target Users

Utility Operations Staff

Billing desk operators who input monthly meter entries and generate official consumer statements without manual mathematical risk.

Residential & Business Consumers

End-users like **Rahul Sharma** who receive transparent, broken-down statements detailing consumption slabs, taxes, and fixed charges.

4. High-Level Features

- Automated Meter Data Validation:** Built-in sanity check ensuring current meter readings are strictly greater than or equal to previous readings, preventing negative bill charges.
- Tiered Progressive Tariff Engine:** Dynamic calculation based on customizable consumption blocks to incentivize

5. Tariff Structure & Calculation Logic

The system implements a four-tier progressive electricity tariff structure. Energy charges are computed sequentially through slab limits:

Consumption Block	Tariff Rate	Calculation Logic Applied
0 – 100 units	₹3.00 / unit	Units within first 100 × ₹3
101 – 200 units	₹5.00 / unit	(100 × ₹3) + (Remaining units up to 200 × ₹5)
201 – 300 units	₹7.00 / unit	(100 × ₹3) + (100 × ₹5) + (Remaining units up to 300 × ₹7)
Above 300 units	₹9.00 / unit	(100 × ₹3) + (100 × ₹5) + (100 × ₹7) + (Remaining units × ₹9)

6. Technical Implementation & Module Mapping

The codebase is structured into cohesive Python scripts designed to perform dedicated functions within the execution pipeline:

Module Name	Primary Responsibility	Key Functions / Logic
calculator.py	Core Tariff Computation Engine	Exports <code>calculate_energy_charge(units)</code> for multi-tier calculations.
bill.py	Bill Formatting & Rendering	Exports <code>display_bill(...)</code> to generate structured consumer invoices.
validation.py	Input Verification	Validates that <code>current_reading ≥ previous_reading</code> prior to calculation.
final.py	Main Execution Orchestrator	Imports core modules, processes customer data, applies taxes, and drives execution.

7. System Validation & Verification Summary

Execution Baseline Test Case

Customer Name : Rahul Sharma Previous Reading : 1250 units | Current Reading : 1495 units
Units Consumed : 245 units [Energy Charge] : (100*3) + (100*5) + (45*7) = ₹1,115.00 [Fixed Charge] : ₹100.00 [Subtotal] : ₹1,215.00 [Tax (5%)] : ₹60.75 -----
----- TOTAL BILL : ₹1,275.75

